Week 1.6 — TriggerFish Setbacks — Steps Back That Move Us Forward

Sunday, July 19, 2026 · 3:00 – 5:30 PM

Class 1.6 was a night of setbacks that turned into the whole lesson. Four of us — Rebecca, Tyler, Frankie, and Isaac — worked the TriggerFish control-box side of the build from 3:00 to 5:30. Rebecca soldered the topside tether lead onto the 8-pin connector, and she did it really well. Then we tried to seat the metal cover — and the eight color-coded conductors were too long to fit inside. Everything had to come apart. Cables trimmed shorter. All eight joints re-soldered. About an hour and a half we won't get back. On the bench test we still couldn't get the LED indicators to read anything but red — polarity is off somewhere and none of us can tell why yet, so Coach John's staying on the TriggerFish manufacturer to sort it out before we power anything up. While the tether work was happening, the rest of the team broke off to install the backplate into the yellow Pelican case. Halfway through, one of the power cables came undone inside a plastic protective sleeve. Re-setting that crimped joint was precision work — no room to force it — and it swallowed another chunk of the night. But by the end we got the tether backplate mounted onto the case and the video wire soldered onto the backplate. Two lessons landed hard: sometimes you take a step backward to take a step forward — that's what perseverance actually looks like. And watch the whole install video before you start work, even when the task feels intuitive. The gotchas live in the parts you'd skip.

4	8	~1.5 hr	1
STUDENTS	CONDUCTORS RE-SOLDERED	SOLDER REWORK	VIDEO WIRE INSTALLED ON BACKPLATE

WHAT WE LEARNED

- Sometimes you take a step backward to take a step forward. Rebecca soldered the topside tether beautifully — and we had to undo every joint because the cables were too long for the cover. Ninety minutes of the same solder done twice. That's not failure — that's what a rework looks like when we catch a problem before it ships to the pool.
- Watch the whole video before you start. Even with a strong intuitive sense of what to do, the video is where the gotchas live — like cable-length-vs-enclosure-space — that intuition doesn't warn you about.
- A crimped joint inside a plastic sleeve is a precision job, not a strength job. When the power cable came apart during the backplate install, force wasn't going to fix it. Slowing down did.
- When bench-test LEDs come back the wrong color, you don't power up and see what happens. You meter the board, you ask the manufacturer, and you don't move on until you understand why.

STUDENT LEARNING OUTCOMES

- First hands-on solder of the topside tether — 8 color-coded conductors mapped to the SeaMATE pinout — including the humbling experience of taking every joint apart and doing it over.
- First TriggerFish backplate mounted into the yellow Pelican control-box housing; first video wire soldered to the backplate.
- Real debugging experience: reversed LED polarity that doesn't yet have an answer — and the discipline to stop, meter, and reach the manufacturer instead of powering through.
- The reflex of watching the full reference video before starting — the difference between a "working" build and a "crafted" one that we named at 1.3.

LOOKING AHEAD

Class 1.7 is Capsule Build & First Splash — the team builds the watertight capsules and heads to the pool for the first time. Every kid in the water. Capsule drop #1 with the towel check. Meanwhile the electronics track keeps chasing the LED-polarity issue with the manufacturer, and the frame group cuts the 45° / 65° / 75° pipe segments for the vertical motors.

DECISIONS MADE

- Watch the full install video before starting the work, every time. Cable-length-vs-enclosure clearance lives in the parts you'd skip.
- Measure cable length against the enclosure BEFORE the final solder. Cable management is a design step, not a cleanup step.
- Do not smoke-test the control board until the TriggerFish manufacturer answers on the reversed LED polarity. Verified beats "probably working."
- Perseverance is the standard: setbacks get named and worked through, not skipped over.

ACTION ITEMS

Follow up with the TriggerFish manufacturer on reversed LED polarity — hold power-up until answer is in
Coach John · ongoing

Standing rule: measure the enclosure clearance BEFORE final solder — "measure the space, not the wire"
Sunday cohort · ongoing

Cut vertical-motor pipe segments at 45° / 65° / 75° using SOHCAHTOA on the length
Frame group · by 1.7

Trade-study depth sensor (analog vs digital) and pick a camera + display path
Sunday cohort · by 1.8

Continue backplate wiring and finish the control-box case assembly
Sunday cohort · by 1.7

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